Direct-Best Reversals in Train-Free Structured Counterfactual Reasoning

Anonymous submission

Abstract

Structured prompts are often treated as a safe train-free way to improve counterfactual reasoning, but a positive single-benchmark result can hide direct-best reversals under a different answer interface or model backend. We test this failure mode by comparing direct prompting, chain-of-thought, ledger, CoIn-style backtracking, a World-State Intervention Scaffold (WSIS), three WSIS ablations, and a repaired adaptive role selector on CounterBench, CLADDER, and fixed-target CRASS. Across 24 Qwen/Qwen2.5-7B-Instruct condition rows and 23,168 scored answers, structure is not uniformly helpful: CounterBench favors backtracking at 0.7083 versus 0.5175 direct accuracy with 301 paired wins and 72 losses, CLADDER peaks at the no-propagation ablation at 0.5556, but CRASS is direct-best at 0.8358. Robustness evidence turns the mixed result into the main finding rather than a caveat: a full Llama key-condition run is direct-best on all three families, Swallow CounterBench is direct-best or tied, and llm-jp reproduces only part of the Qwen direction. A repaired adaptive selector is auditable but fails as a rescue mechanism, with zero positive family/backend slices above threshold and three negative slices. The paper's contribution is a negative evaluation protocol: structured counterfactual prompts must be judged by paired direct baselines, direct-best reversals, answer-interface stress tests, and backend sensitivity before being reported as a method.

Introduction

Counterfactual questions are difficult for language models because they require two operations that fluent continuation can blur together: changing the facts that follow from an intervention and preserving the facts that should remain unchanged. This distinction is central to formal accounts of possible worlds, treatment effects, and structural interventions (Lewis 1973; Rubin 1974; Pearl 2009; Pearl and Mackenzie 2018). Recent causal and counterfactual benchmarks translate the operation into natural-language questions with scorable final answers (Jin et al. 2023; Chen et al. 2025), but the resulting accuracy number does not reveal whether the model revised a world state, followed a superficial cue, or produced a plausible trace that happened to end correctly.

A common response is to add reasoning structure at inference time. Chain-of-thought prompting makes intermediate text visible (Wei et al. 2022), and causal-reasoning prompts can ask for variables, interventions, or graph-like state (Kim et al. 2024). The risk is that these prompts can improve one benchmark while silently losing to direct prompting on another family or backend. The useful comparison is therefore falsification-oriented: which structured controls beat direct on the same examples, which controls reverse under a different answer interface, and whether a selector can recover a stable policy after those reversals are visible.

We evaluate that question with a train-free World-State Intervention Scaffold (WSIS). WSIS is a prompt controller around a fixed language-model backend: it asks the model to write a state ledger, perform intervention surgery, propagate affected consequences when the prompt supports them, preserve invariants, and then emit the benchmark answer. It uses only public benchmark fields and does not train on labels, inspect gold rationales, or use hidden graph fields. We also test ablations and a repaired adaptive role selector, because a plausible response to heterogeneity is to choose a role policy per example. The paper's scientific object is whether those controls survive direct-baseline, answer-interface, and backend stress tests. Figure 1 summarizes the evaluated pipeline.

The completed run covers CounterBench (Chen et al. 2025), CLADDER (Jin et al. 2023), and repaired fixed-target CRASS (Frohberg and Binder 2021). It falsifies a simple method story. CounterBench gives the strongest positive row, with CoIn-style backtracking at 0.7083 accuracy versus 0.5175 for direct prompting and 301 paired wins against 72 losses. CLADDER gives a smaller structured-family result, where the no-propagation ablation is highest at 0.5556 and full WSIS is 0.5527. CRASS reverses the conclusion under a fixed-target answer interface: direct prompting is best at 0.8358, above CoT at 0.8102 and full WSIS at 0.8029.

Robustness and trace evidence determine the claim boundary. A repaired llm-jp diagnostic reproduces the CounterBench backtracking direction, 0.4950 versus 0.3558 direct, but its CLADDER and CRASS positives are small or non-significant. Llama and Swallow do not reproduce the Qwen CounterBench ordering, and a full Llama three-family key-condition run is direct-best on CounterBench, CLADDER, and CRASS. Trace-text validation over 41 paired method-versus-direct comparisons adds a process-facing diagnostic: visible intervention and propagation controls occur in some positive rows, but the trace evidence remains

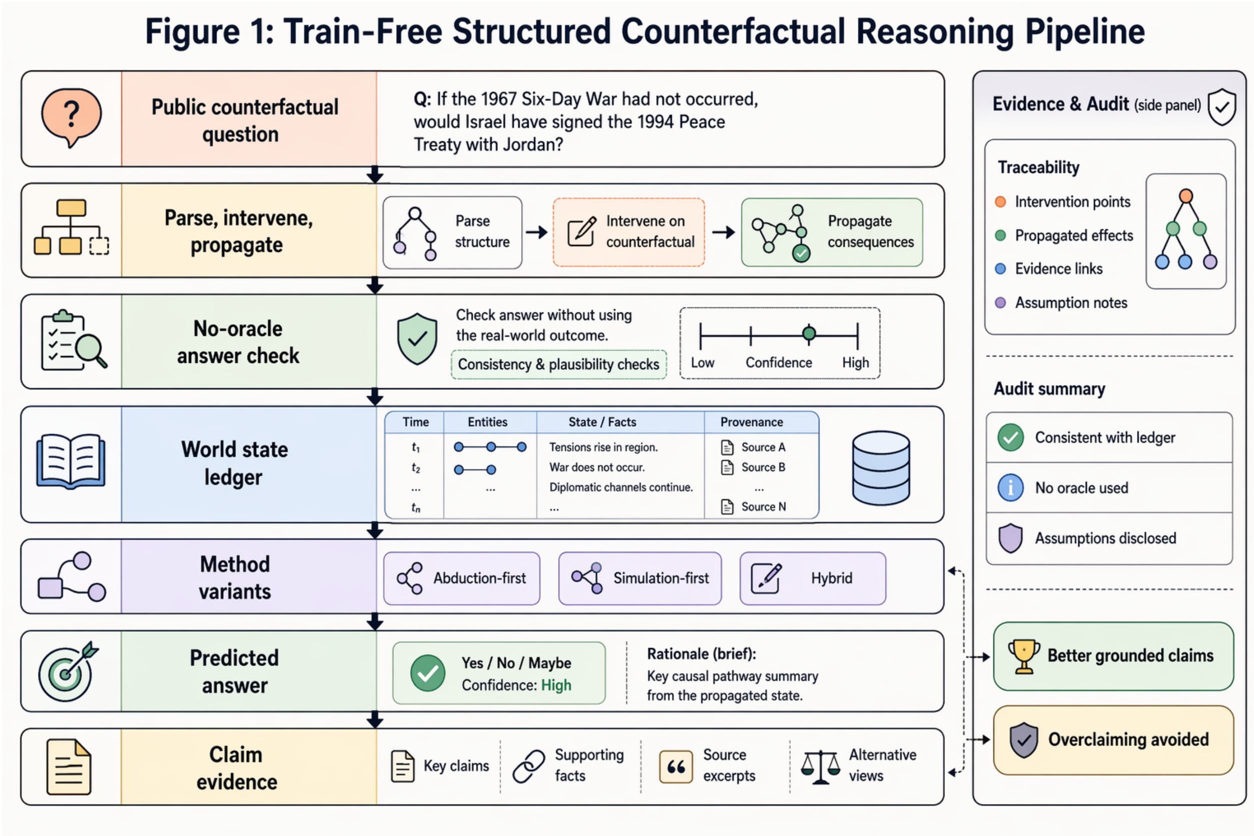

Figure 1: Method pipeline: public counterfactual questions are transformed into a world-state ledger, updated by train-free intervention and propagation steps, and interpreted as scoped family-specific claims.

conditional rather than a proof of a uniform mechanism. A repaired adaptive role selector sharpens the negative result. It emits parseable policies, but across two instruction-tuned backends and three families it has zero direct-comparison slices above the positive threshold and three negative slices.

The contribution is therefore a negative evaluation protocol for train-free structured counterfactual prompting. The paper defines WSIS roles, evaluates them against direct, chain-of-thought, prompt-ledger, and backtracking controls, and reports the ablations, robustness rows, adaptive-selection failure, and trace diagnostics that a positive method paper would usually treat as secondary. The actionable conclusion is that a structured-prompt claim should not be reported from a single positive family: it should survive paired direct baselines, direct-best reversal checks, answer-interface stress tests, and backend sensitivity analysis.

Related Work

Formal counterfactual reasoning builds on theories of possible worlds, treatment effects, and structural causal models (Lewis 1973; Rubin 1974; Pearl 2009; Pearl and Mackenzie 2018). These foundations motivate evaluations where the answer depends on intervention semantics rather than topical plausibility: changing the intervened variable, propagating only warranted consequences, and leaving invariant facts alone.

Language-model causal reasoning benchmarks have developed along several lines. CRASS tests counterfactual conditionals in natural language (Frohberg and Binder 2021). CLADDER grounds natural-language causal questions in symbolic causal queries and reports final-answer and step-wise behavior (Jin et al. 2023). CounterBench focuses on counterfactual reasoning over constructed worlds and evaluates CoIn-style iterative backtracking (Chen et al. 2025). Broader reviews argue that causal-reasoning benchmarks must distinguish formal reasoning from pattern recall (Yang et al. 2024). Adjacent counterfactual-evaluation work extends this caution to axiomatic image counterfactuals, visual-linguistic causal tasks, and visual causal benchmarks (Monteiro et al. 2023; Liu et al. 2023; Melistas et al. 2024; Komanduri, Bhaila, and Wu 2025); recent LLM studies similarly decompose causal or counterfactual competence before making broad claims (Kıcıman et al. 2023; Yang et al. 2025; Hase and Potts 2026).

Structured prompting and reasoning traces are often used to make model outputs more inspectable. Chain-of-thought prompting is a general instance of this strategy (Wei et al. 2022; Kojima et al. 2022; Wang et al. 2022; Zhou et al. 2022). Program- and action-oriented variants separate reasoning text from operations or environment feedback (Gao et al. 2022; Chen et al. 2022; Yao et al. 2022; Shinn et al. 2023); search, feedback, and alignment variants add control over intermediate states (Yao et al. 2023; Madaan et al. 2023; Besta et al. 2023; Wang et al. 2023). Causal-reasoning variants add task-specific structure or external representations (Kim et al. 2024). The scaffold studied here is narrower: it tests train-free state and propagation roles under one protocol, not whether a new trained model has learned causality.

Finally, the evaluation design follows benchmark-audit work that treats aggregate accuracy as insufficient without behavioral slices, compositional controls, and task-family provenance (Ribeiro et al. 2020; Tafjord, Dalvi Mishra, and Clark 2020; Press et al. 2022). General LLM benchmark reports reinforce why protocol, scorer, and task selection details matter for reasoning claims (Hendrycks et al. 2020; Lin, Hilton, and Evans 2021; Cobbe et al. 2021; Zheng et al. 2023).

The closest methodological comparison is therefore not a new causal model, but a set of inference-time controls that allocate different roles to the same language model. Direct prompting tests whether the model can answer from the question alone. Chain-of-thought tests whether unconstrained rationale text is enough. Prompt ledger separates explicit state recording from intervention and propagation. Backtracking tests whether revisiting intermediate states changes the final answer. The structured symbolic and ablated variants test whether the full parse-intervene-propagate-invariant sequence is better than removing one of its roles. This comparison is intentionally narrower than claims about causal competence, but it is the right granularity for measuring whether a train-free scaffold changes benchmark behavior.

Benchmark Families

CounterBench is the clearest fit for the paper's main positive result because it evaluates counterfactual reasoning in constructed worlds and includes reasoning types that require more than direct factual recall (Chen et al. 2025). In this project's completed run, CounterBench has 1,200 scored rows for each evaluated condition. That balanced per-condition shape makes paired comparisons against direct prompting straightforward: every method condition can be compared against the same examples, and the resulting win/loss/tie counts are easy to inspect. The benchmark is also aligned with the method design because CoIn-style backtracking is one of the evaluated conditions, so the paper can compare a search-like reasoning control against direct, chain-of-thought, ledger, and structured-symbolic variants.

CLADDER provides a complementary stress test. It asks causal questions grounded in symbolic causal queries and natural-language verbalizations (Jin et al. 2023). In the completed run, CLADDER contributes 1,422 scored rows per condition. The benchmark is useful for this paper because it separates the final-answer question from the idea of a reasoning trace: a condition can produce a final yes/no answer while exposing more or less structured detail about graph parsing, formalization, intervention, and arithmetic. This makes CLADDER a natural place to test whether the full structured symbolic scaffold beats ablations. The observed answer is nuanced: structured variants improve over direct, but the no-propagation ablation is the highest tested condition.

CRASS is now part of the executed evidence rather than a boundary-only source. The repaired fixed-target adapter exposes public answer choices and scores choice-letter predictions against scorer-only labels. In the completed run, CRASS contributes 274 examples per condition, or 2,192 scored rows across the same eight conditions used for the other families. It strengthens benchmark coverage but not the proposed-method claim: direct prompting is the family winner at 0.8358, while all structured and prompt variants are lower.

The three-family matrix determines the paper's claim vocabulary. The paper uses "benchmark family" rather than "task" when it reports aggregate results because CounterBench, CLADDER, and CRASS differ in row construction, causal-query framing, and answer interface. It uses "condition" rather than "model" for direct, chain-of-thought, ledger, CoIn backtracking, structured symbolic, and ablation rows because the evaluated model/backend is held fixed. It uses "downgrade" for negative-delta conditions because those observations prevent stronger claims but remain scientifically useful.

This benchmark framing is also why the paper does not pool the three families into a single headline average. The current body is organized around what was actually run: three source-backed families, eight conditions per family, and a family-specific interpretation in which CounterBench favors backtracking, CLADDER favors no propagation, and CRASS favors direct prompting.

Method

The evaluated framework is the World-State Intervention Scaffold (WSIS), a paper-facing prompt controller that converts a public benchmark question into an answer through five stages. The parser stage asks for variables, relationships, observations, and the requested counterfactual target. The intervention stage rewrites the affected variable to its hypothetical value. The propagation stage updates dependent facts when the prompt exposes enough public structure for such an update. The invariant stage preserves facts that should not change under the intervention. A final extraction stage emits the answer in the benchmark's expected form.

The method is evaluated through variants rather than claimed as a single monolithic solution. Direct prompting and chain-of-thought are prompting baselines. Prompt ledger asks for explicit state without the full scaffold. CoIn-style backtracking asks the model to revisit intermediate reasoning states. The full WSIS condition asks for parse, intervention, propagation, invariant preservation, and answer extraction. Three ablations remove propagation, invariant preservation, or intervention surgery. These ablations are important because the run shows that removing a component can improve one benchmark family, so the paper cannot claim that the full structured condition dominates across the evaluated families.

We also run an adaptive role-selection diagnostic after observing the family and backend reversals. The selector is a prompt-only controller: before answering, the model must emit a `POLICY:` line choosing direct-only, backtracking, full WSIS, or an ablated structured policy, and the runner records the selected policy before scoring the final answer. This diagnostic is not part of the 24-row primary Qwen matrix. It is a rescue test for the heterogeneity finding: if family/backend dependence is merely a selection problem, an auditable policy line should recover positive slices. The powered selector run uses two instruction-tuned backends, Qwen/Qwen2.5-7B-Instruct and NousResearch/Meta-Llama-3.1-8B-Instruct, with 300 CounterBench rows, 300 CLADDER rows, and all 274 CRASS rows per backend. We treat a family/backend slice as positive only if its paired delta versus direct is greater than +0.02, and negative if it is less than -0.02.

The method uses no training in this paper. It also does not use hidden labels, gold rationales, or benchmark-private graph fields for inference. Scoring is performed after generation by the locked benchmark scorer. This separation is what lets the paper report both wins and failures without converting diagnostic signals into oracle assistance.

Experiments

The full run covers three benchmark families. CounterBench contributes 1,200 scored rows per method condition, CLADDER contributes 1,422 scored rows per condition, and repaired fixed-target CRASS contributes 274 scored rows per condition. Across eight conditions and three families, the evaluation contains 24 condition rows and 23,168 scored rows. The primary full-matrix backend is Qwen/Qwen2.5-7B-Instruct with deterministic decoding temperature 0.0 and top-p 1.0. CounterBench and CLADDER use a 512-token generation cap; CRASS uses a 16-token fixed-target answer cap.

The primary metric is binary answer accuracy. For each cell, the analysis reports Wilson confidence intervals. For each non-direct condition, the analysis also reports paired wins, losses, ties, mean delta, and a two-sided sign test excluding ties. All reported table values are computed from the same scored predictions, so aggregate accuracies, paired comparisons, and ablation summaries refer to the same underlying examples.

The experiment design deliberately keeps model and method variation separate. The primary matrix uses the same evaluated model/backend and decoding settings, so those comparisons are method-condition comparisons under a fixed protocol. This matters for interpretation: within the Qwen full matrix, a gain for CoIn-style backtracking on CounterBench is not a statement that a different model would show the same ordering, and a CLADDER ablation ordering is not evidence that propagation is generally harmful. The single-run full matrix gives a controlled comparison among conditions, but it leaves multi-seed and cross-hardware questions open.

To probe that boundary, we also run model-variation robustness diagnostics. The first diagnostic uses NousResearch/Meta-Llama-3.1-8B-Instruct on the first 100 rows of each family for direct, CoIn backtracking, structured symbolic, and no propagation. Because the initial 300-row CounterBench backtracking extension produced many missing-final-answer rows, we add corrected compact-stop CounterBench slices: one on the first 300 examples for diagnosis and one on all 1,200 CounterBench tasks. We then complete a full Llama key-condition run over CounterBench, CLADDER, and CRASS with the same direct, backtracking, structured symbolic, and no-propagation conditions. This full Llama run contributes 11,584 scored rows and is direct-best on all three families, so it is negative robustness evidence for broad method superiority. We then add tokyotech-llm/Llama-3.1-Swallow-8B-Instruct-v0.3 diagnostics: a full CounterBench direct/backtracking compact-stop slice, plus full CLADDER and CRASS direct, backtracking, structured-symbolic, and no-propagation slices. Finally, we add a full llm-jp-3-7.2B-Instruct diagnostic on CounterBench, CLADDER, and CRASS for direct, CoIn backtracking, structured symbolic, and no propagation. The first llm-jp structured-symbolic CounterBench/CLADDER cells produced many empty outputs, so we reran only those two affected cells with a compact final-answer contract; the repaired cells have 0 empty outputs over 2,622 rows under the same public-input scoring protocol. These runs are reported as robustness diagnostics rather than merged into the full Qwen matrix.

Each benchmark family is evaluated as a paired matrix. Direct prompting is the within-family baseline. Every other condition is run on the same example ids and then compared against direct with a sign test over non-tied examples. This paired design is important because aggregate accuracy alone can hide whether a condition changes the same examples or

Setup item	Reader-facing protocol
Benchmarks	CounterBench has 1,200 rows per condition; CLADDER has 1,422 rows per condition; repaired fixed-target CRASS has 274 rows per condition.
System	One fixed backend, Qwen/Qwen2.5-7B-Instruct, with direct, CoT, ledger, backtracking, full WSIS, and three ablation prompts.
Budget	Deterministic decoding with temperature 0.0 and top-p 1.0; 512-token caps for CounterBench/CLADDER and 16-token caps for CRASS.
Scorer	Locked final-answer scorer; binary accuracy with Wilson intervals; paired sign tests against direct within each family.
Readout	Report the 24-row matrix first, then interpret family-specific paired gains, losses, and ablation orderings.
Adaptive rescue diagnostic	Separate two-backend policy-line selector: 300 CounterBench rows, 300 CLADDER rows, and all 274 CRASS rows per backend; positive if paired delta vs direct is $> +0.02$.

Table 1: Evaluation setup: the same backend, final-answer scorer, and paired-comparison protocol are used for the 24 condition rows, with CRASS using its repaired fixed-target answer cap.

only shifts a small number of marginal cases. It also keeps negative evidence visible: a condition with an accuracy near direct can still have a negative paired delta, and such rows should not be described as improvements.

The CRASS direct-best rows in Table 2 are the key answer-interface reversal that keeps the main interpretation family-specific rather than uniform.

Claim Calibration

The results require calibration because three objects can otherwise be confused: a benchmark family, an inference-time condition, and a broader causal reasoning capability. The safest reading is family-specific. CounterBench supports a clear backtracking result. CLADDER supports a smaller structured-variant result with a different best condition. CRASS supports executed coverage but gives the best score to direct prompting. No family justifies a statement that the full structured symbolic condition is uniformly best.

On CounterBench, the strongest result is CoIn-style backtracking. It reaches 0.7083 accuracy against 0.5175 for direct prompting. The paired delta is +0.1908, with 301 wins and 72 losses. The no-invariant ablation also improves over direct at 0.6258 accuracy and +0.1083 paired delta. These two rows show that explicit state revision and some structured controls can help this benchmark family. They do not show that every scaffold component is beneficial.

CLADDER gives a different pattern. The full structured symbolic condition improves over direct, reaching 0.5527 accuracy with a +0.0485 paired delta. The no-propagation ablation is slightly higher at 0.5556. Both observations are needed. Reporting only the full condition would hide the ablation that wins the family. Reporting only the ablation would hide that the full structured variant also improves over direct.

The negative rows are part of the result rather than a reporting inconvenience. CounterBench prompt ledger and no propagation have negative paired deltas against direct, so they are not improvement evidence. CRASS is positive executed coverage, but every non-direct CRASS condition is below direct: CoT is 0.8102, full WSIS is 0.8029, and no propagation is also 0.8029 against direct at 0.8358. These boundaries keep the paper's conclusion tied to answer-accuracy improvements on named families rather than to a broad claim about solved counterfactual reasoning.

Table 3 adds the boundary needed for the rescoped claim. The full corrected Llama key-condition run is direct-best on all three families: direct beats backtracking on CounterBench by 0.0467, beats backtracking and structured symbolic on CLADDER, and remains highest on CRASS. Swallow CounterBench is also direct-best or tied, with its no-propagation row confounded by missing final-answer format. In contrast, llm-jp reproduces the Qwen CounterBench direction: backtracking is 0.4950 versus 0.3558 direct, and repaired structured prompting is 0.4917. Its CLADDER and CRASS positives are small or non-significant, as are the Swallow CLADDER/CRASS positives. These rows justify a heterogeneity/null-results framing rather than a cross-backend method-dominance claim.

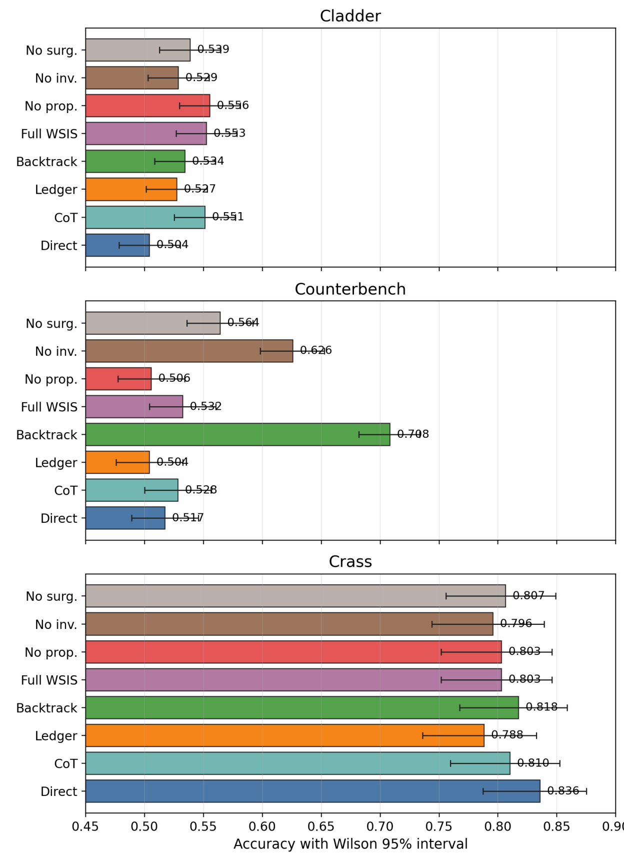

Figure 2: Accuracy with Wilson intervals: CounterBench peaks at 0.7083 for CoIn backtracking, while CLADDER peaks at 0.5556 for no propagation.

Results and Discussion

Figure 2 gives the accuracy profile with Wilson intervals. The plot shows which condition has the highest aggregate score within each family.

The paired tests ask a separate question. They measure whether a condition changes the same examples relative to direct prompting. The paper needs both views because the aggregate winner and the strongest paired movement need not be the same row.

The main family-specific findings are in Table 2. CounterBench's clearest positive row is CoIn backtracking. It reaches 0.7083 accuracy versus 0.5175 for direct prompting. The paired comparison has 301 wins, 72 losses, and 827 ties. Its mean delta is +0.1908 ($p = 2.11 \times 10^{-34}$). On the same family, no invariant reaches 0.6258 accuracy and +0.1083 paired delta. The full structured symbolic row reaches 0.5325 and is not a sign-test-safe win over direct.

CLADDER produces a more cautious result. The full structured symbolic condition improves over direct at 0.5527 accuracy. Its paired delta is +0.0485, with 163 wins, 94 losses, and $p = 1.99 \times 10^{-5}$. It is not the highest tested condition. The no-propagation ablation reaches 0.5556 accuracy, slightly above the full structured symbolic variant.

Benchmark	Backend/budget	Method	Rows	Accuracy	Delta	Wins–losses	Sign-test
CounterBench	Qwen2.5-7B; 512 tok.	Direct	1200	0.5175	–	–	baseline
CounterBench	Qwen2.5-7B; 512 tok.	CoT	1200	0.5283	+0.0108	64–51	n.s.
CounterBench	Qwen2.5-7B; 512 tok.	Prompt ledger	1200	0.5042	-0.0133	33–49	downgrade
CounterBench	Qwen2.5-7B; 512 tok.	CoIn backtracking	1200	**0.7083**	+0.1908	301–72	2.11×10^{-34}
CounterBench	Qwen2.5-7B; 512 tok.	Structured symbolic	1200	0.5325	+0.0150	63–45	n.s.
CounterBench	Qwen2.5-7B; 512 tok.	No propagation	1200	0.5058	-0.0117	41–55	downgrade
CounterBench	Qwen2.5-7B; 512 tok.	No invariant	1200	0.6258	+0.1083	209–79	1.02×10^{-14}
CounterBench	Qwen2.5-7B; 512 tok.	No surgery	1200	0.5642	+0.0467	148–92	3.64×10^{-4}
CLADDER	Qwen2.5-7B; 512 tok.	Direct	1422	0.5042	–	–	baseline
CLADDER	Qwen2.5-7B; 512 tok.	CoT	1422	0.5513	+0.0471	109–42	4.76×10^{-8}
CLADDER	Qwen2.5-7B; 512 tok.	Prompt ledger	1422	0.5274	+0.0232	146–113	4.66×10^{-2}
CLADDER	Qwen2.5-7B; 512 tok.	CoIn backtracking	1422	0.5345	+0.0302	117–74	2.29×10^{-3}
CLADDER	Qwen2.5-7B; 512 tok.	Structured symbolic	1422	0.5527	+0.0485	163–94	1.99×10^{-5}
CLADDER	Qwen2.5-7B; 512 tok.	No propagation	1422	**0.5556**	+0.0513	197–124	5.47×10^{-5}
CLADDER	Qwen2.5-7B; 512 tok.	No invariant	1422	0.5288	+0.0246	106–71	1.04×10^{-2}
CLADDER	Qwen2.5-7B; 512 tok.	No surgery	1422	0.5387	+0.0345	139–90	1.46×10^{-3}
CRASS	Qwen2.5-7B; 16 tok.	Direct	274	**0.8358**	–	–	baseline
CRASS	Qwen2.5-7B; 16 tok.	CoT	274	0.8102	-0.0255	0–7	1.56×10^{-2}
CRASS	Qwen2.5-7B; 16 tok.	Prompt ledger	274	0.7883	-0.0474	5–18	1.06×10^{-2}
CRASS	Qwen2.5-7B; 16 tok.	CoIn backtracking	274	0.8175	-0.0182	4–9	n.s.
CRASS	Qwen2.5-7B; 16 tok.	Structured symbolic	274	0.8029	-0.0328	4–13	4.90×10^{-2}
CRASS	Qwen2.5-7B; 16 tok.	No propagation	274	0.8029	-0.0328	4–13	4.90×10^{-2}
CRASS	Qwen2.5-7B; 16 tok.	No invariant	274	0.7956	-0.0401	4–15	1.92×10^{-2}
CRASS	Qwen2.5-7B; 16 tok.	No surgery	274	0.8066	-0.0292	4–12	n.s.

Table 2: Main results: all 24 condition rows use Qwen/Qwen2.5-7B-Instruct with deterministic decoding. Bold marks family winners: CounterBench CoIn at 0.7083, CLADDER no propagation at 0.5556, and CRASS direct at 0.8358.

That small ordering difference is decisive for claim calibration. CLADDER supports structured conditions in this run, but not a uniform full-method win.

The strongest paired tests are reported in prose above, while Figure 3 shows the full paired-delta plot. CounterBench prompt ledger and no propagation remain downgraded because both have negative paired deltas. The result is not that every explicit structure helps; particular controls help particular benchmark families.

The direct baseline remains the anchor because every condition uses the same public question text and final-answer scorer. Chain-of-thought and prompt-ledger runs separate rationale text from explicit state recording. Neither transfers uniformly across families, so Table 2 reports accuracy, paired delta, win/loss counts, and sign-test evidence together.

The paired-delta view in Figure 3 prevents a single aggregate reading. CounterBench CoIn has the largest movement at +0.1908. Full structured symbolic is only +0.0150, and prompt ledger/no propagation are negative. CLADDER has the opposite shape: every non-direct row is positive, the spread is narrow, and no propagation is highest. This is why the paper does not collapse the three families into one average.

A separate trace-text validation asks whether generated traces expose the controls that the prompts request. It re-reads 52 scored-row source files and forms 41 paired method-versus-direct comparisons from the primary and robustness runs, then stratifies the paired rows into 608 feature strata and 198 feature-lift rows. The aggregate result is again conditional rather than uniform: 15 paired deltas are above +0.02, 14 are below -0.02, and high intervention-plus-propagation trace coverage appears in 3 positive pairs and 0 negative pairs. At the task level, Qwen CounterBench backtracking rows that mention propagation have +0.2052 accuracy delta with 220 wrong-to-right changes and 47 right-to-wrong changes, but Llama CounterBench no-propagation rows that mention intervention have -0.4965 delta with 3 wrong-to-right changes and 144 right-to-wrong changes. This supports a mechanism diagnostic–visible controls can coincide with gains or regressions–but it does not prove an internal causal mechanism or multi-seed robustness.

A complementary role-ablation audit asks a more interventional question: what happens when prompted scaffold roles are removed in executed cells? It covers 54 scored cells and 40 paired comparisons across the three benchmark families and four model slices. The result is stronger than lexical trace matching because the compared conditions actually remove or restore prompt roles, but it remains mixed. Backtracking versus direct has four positive, three negative, and three near-null cells. Full structured prompting versus direct has three positive, five negative, and four near-null cells.

Slice	Family	Key comparison	Rows	Accuracy	Delta	Claim status
Llama	CounterBench	Direct vs backtrack	1200	0.6325 vs 0.5858	-0.0467	direct-best
Llama	CLADDER	Direct vs structured	1422	0.6013 vs 0.5520	-0.0492	direct-best
Llama	CRASS	Direct vs structured	274	0.8321 vs 0.8285	-0.0036	direct-best
Swallow	CounterBench	Direct vs backtrack	1200	0.5067 vs 0.4867	-0.0200	direct-best
Swallow	CounterBench	Direct vs structured	1200	0.5067 vs 0.5067	0.0000	tied
Swallow	CounterBench	Direct vs no propagation	1200	0.5067 vs 0.2825	-0.2242	parse-confounded drop
llm-jp	CounterBench	Direct vs backtrack	1200	0.3558 vs 0.4950	+0.1392	backtracking positive
llm-jp	CounterBench	Direct vs repaired structured	1200	0.3558 vs 0.4917	+0.1358	repaired positive
llm-jp	CLADDER	Direct vs repaired structured	1422	0.5056 vs 0.5415	+0.0359	narrow; n.s.
llm-jp	CRASS	Direct vs backtrack	274	0.5803 vs 0.6095	+0.0292	small; n.s.
Swallow	CLADDER/CRASS	Best small positives	1422/274	0.5000/0.7518	+0.0183/+0.0219	small; n.s.

Table 3: Robustness diagnostics: compact body summary of the model-variation evidence. Detailed paired counts and output-format breakdowns are reported in the appendix and supplementary material.

Propagation-role comparisons are the most active signal, with six positives, two negatives, and four near-null cells, including large positive no-propagation drops for Llama and Swallow but large negative original llm-jp structured cells. The audit therefore supports a prompt-level mechanism diagnostic–roles can causally change final answers–not a stable positive mechanism claim.

The repaired adaptive role-selection run tests the obvious rescue strategy: instead of committing to one structured prompt, ask the model to choose a policy for each example and then execute that policy. This selector is auditable: it produces parseable policy lines for every powered row. The result is still negative as a method path. Appendix Table 4 reports the six direct comparisons: across two instruction-tuned backends and three families, it has zero direct-comparison slices above the positive threshold, three negative slices, and no unparsed policy rows. The selector therefore converts heterogeneity into an explicit negative result: recognizing that roles differ by family is not enough unless the policy-selection signal itself is reliable.

Analysis and Failure Modes

The ablations are the main guardrail against overclaiming. On CounterBench, removing invariant preservation improves over the full structured symbolic condition by +0.0933 accuracy, and removing intervention surgery improves by +0.0317. Removing propagation decreases accuracy. CLADDER has a different pattern: no propagation is slightly above the full condition, while no invariant and no surgery are below it. On CRASS, direct is best and every structured ablation remains below direct. Figure 4 summarizes these family-specific directions.

CRASS is used as executed coverage, but it is a negative method-control result. The repaired fixed-target run contributes 2,192 scored rows with non-constant binary scores and choice-letter outputs. Direct prompting is best at 0.8358. CoT reaches 0.8102, CoIn backtracking reaches 0.8175, full WSIS reaches 0.8029, and all structured or prompt variants are below direct. In this paper, CRASS therefore broadens benchmark coverage while preventing a proposed-method win claim.

The failure taxonomy is also modest. The available annotations describe final-answer format quality and coarse intervention or propagation behavior, but they do not provide multi-seed dispersion or deeper causal trace metrics. The Llama, Swallow, and llm-jp diagnostics add useful model-variation checks, including a full Llama three-family key-condition run, full CounterBench slices, Swallow CLADDER/CRASS slices, and repaired llm-jp three-family key-condition slices. They are mixed for the broadest interpretation of the CounterBench result: Qwen and llm-jp support backtracking, Llama and Swallow do not, Llama is direct-best on all three full key-family slices, and the positive Swallow and llm-jp CLADDER/CRASS deltas are small or non-significant. Accordingly, this paper does not claim robustness across models, seeds, or benchmark families beyond the reported family-specific evidence.

The residual error rates remain large enough to shape the interpretation. The best CounterBench condition still leaves a 0.2917 error rate, and the best CLADDER condition leaves a 0.4444 error rate; the full residual-error plot is included in Appendix Figure 5. These values are incompatible with any claim that the scaffold solves formal counterfactual reasoning. They are, however, compatible with a narrower claim that explicit backtracking and structured controls can change measured behavior under some benchmark families.

The same caution applies to the diagnostic flags and trace text. Some structured outputs did not expose enough public propagation detail for variable-level scoring, and baseline conditions do not expose propagation ledgers at all. The stratified trace validation therefore uses lexical evidence of intervention, propagation, invariants, backtracking, and numbered steps as a diagnostic layer over the scored answers. Its strongest positive strata include CounterBench backtracking on Qwen, but its strongest negative strata include Llama no-propagation rows where visible intervention language accompanies many right-to-wrong flips. The

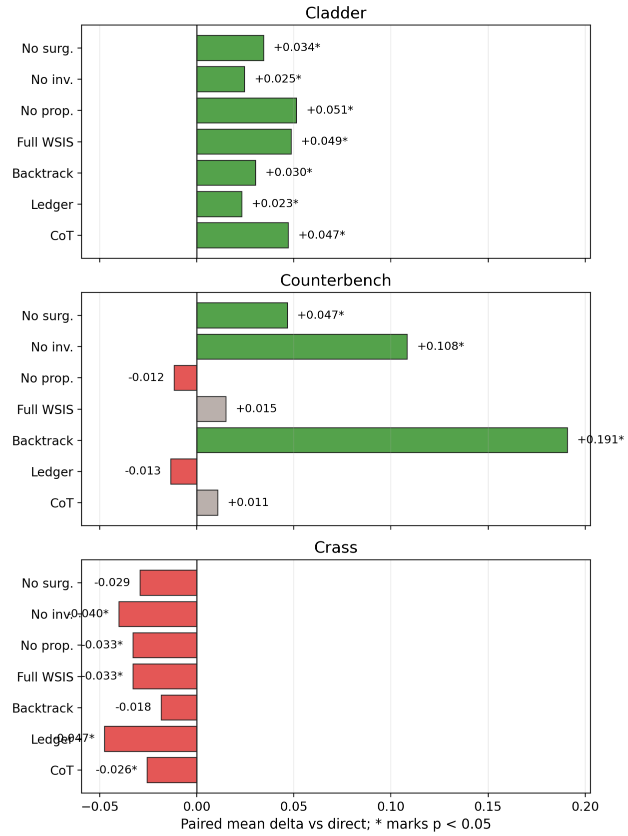

Figure 3: Paired deltas versus direct: CounterBench CoIn has the largest supported gain at +0.1908; prompt ledger and no propagation are downgraded.

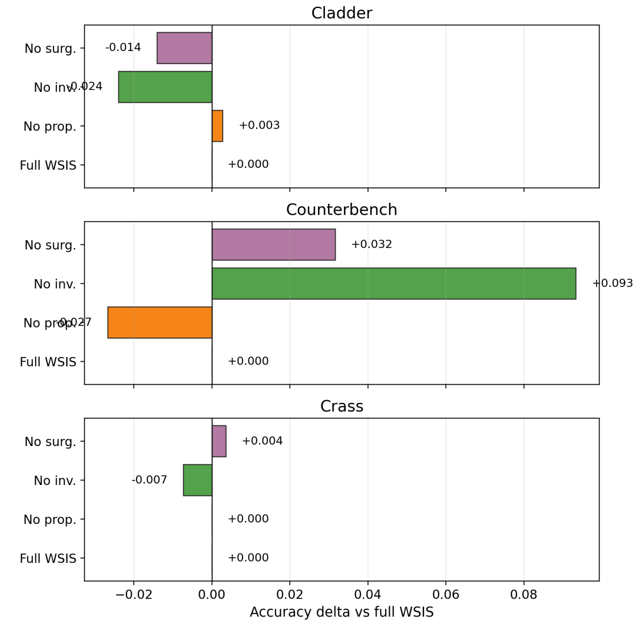

Figure 4: Ablations reject a uniform full-method claim: no invariant leads CounterBench, no propagation leads CLADDER, and CRASS direct remains best.

central measured object remains final-answer accuracy with paired comparisons; the trace features explain when controls are visible in text, not whether the model's hidden computation is causally correct.

The role-ablation audit reaches the same boundary through executed interventions rather than text features. Removing propagation is often damaging relative to full structured prompting on Llama CounterBench and CLADDER and on Swallow CounterBench, which shows that the propagation instruction can matter. However, the same role comparison is near-null on Qwen CLADDER and CRASS and is negative in the original llm-jp CounterBench and CLADDER structured cells before the compact-output repair. This prevents the paper from presenting propagation as a robust mechanism; it is a role whose value depends on backend, answer interface, and output contract.

Implications for Structured Reasoning

The main implication is a falsification rule for train-free structured reasoning claims. A scaffold that wins on one benchmark family has not yet shown that structure improves counterfactual reasoning; it has only shown that one prompting control helped one answer interface under one backend. In this study, that distinction matters. CounterBench supports backtracking for Qwen and llm-jp, CLADDER supports smaller structured effects, CRASS is direct-best, and Llama is direct-best on all three key-condition families. The adaptive selector then fails to recover a stable policy from those differences.

This does not imply that structure is cosmetic. Qwen CounterBench backtracking changes many paired examples, Qwen CLADDER structured variants improve over direct, trace validation finds positive intervention-plus-propagation strata, and role ablations show that removing prompted roles can change final answers. The negative result is narrower and more useful: visible structure is an unstable control surface. Its value depends on benchmark family, answer interface, backend, and output contract, and those dependencies can reverse the winner back to direct prompting.

The supported recommendation is therefore procedural rather than method promotional. Structured counterfactual-prompt papers should report the full condition matrix, keep direct prompting as a paired baseline, include direct-best reversal checks, test at least one fixed-target answer interface, and preserve robustness rows that contradict the preferred mechanism story. Replication across additional backends, seeds, and decoding policies is the next empirical step before any stable method recommendation.

Conclusion

This paper began from a plausible train-free method hypothesis: explicit world-state roles, backtracking, and role selection should improve counterfactual final answers. The completed evidence does not support that hypothesis as a general method claim. CounterBench favors CoIn/backtracking under Qwen, CLADDER shows smaller structured evidence with an ablation-best row, and CRASS gives the highest score to direct prompting in the Qwen matrix. Llama, Swallow, and llm-jp diagnostics show that the strongest Qwen ordering is backend-sensitive rather than robust across backends, and the repaired adaptive selector fails as a rescue mechanism.

The contribution is the resulting negative evaluation: direct-best reversals are not nuisance rows to average away, but the evidence that determines whether a structured prompt is a method or a family-specific control. For counterfactual reasoning, the safe claim is not that train-free structure solves the task. It is that structure can change final answers, while paired direct baselines, answer-interface stress tests, backend checks, and failed policy-selection attempts are necessary before a positive scaffold story is credible.

Scope, Limitations, and Ethics

The first limitation is benchmark coverage and heterogeneity. CounterBench, CLADDER, and CRASS ask related causal or counterfactual questions, but they do not exhaust the space, and CRASS is valuable precisely because it keeps direct prompting ahead.

The second limitation is model coverage. The completed full matrix uses one backend and one deterministic decoding policy. The Llama, Swallow, and llm-jp diagnostics add model-variation checks, but they do not certify robustness: after removing a missing-final-answer confound, Llama direct still exceeds Llama backtracking on CounterBench, Swallow shows the same CounterBench ordering, and llm-jp shows the opposite ordering in favor of backtracking. The repaired llm-jp CLADDER structured cell clears an output-contract confound and has a higher point estimate than direct, but the paired test is not significant; Swallow CLADDER/CRASS and llm-jp CRASS likewise remain non-significant.

The third limitation is trace observability. WSIS asks for ledgers and propagation decisions, and the trace validation measures whether those controls are visible in generated text across 41 paired comparisons. That evidence is useful but lexical: it does not intervene on hidden states, and it cannot certify that each intermediate variable update is causally valid. A stronger version would add trace-level rubrics or causal trace interventions that apply to all conditions without giving any condition private labels or oracle graphs. The new prompt-role ablation audit is stronger than lexical trace matching but still operates at the prompt level, not at the model-state level.

These boundaries also determine how the table values should be reused. The Qwen full matrix is suitable for comparing the tested prompting conditions under the stated backend and decoding policy, but it is not a leaderboard score for counterfactual reasoning systems. A later study that changes the backend, sampling policy, scorer interface, or benchmark subset should rerun the paired comparisons rather than treating the present deltas as transferable constants.

The Llama, Swallow, and llm-jp diagnostics should be read the same way. Their value is not that three additional slices settle robustness; it is that they falsify any single-backend robustness sentence. The full corrected compact-stop slices show that Llama and Swallow CounterBench backtracking non-reproduction is not solely a missing-final-line artifact, while the repaired llm-jp diagnostic shows that the Qwen direction can reproduce on another backend and that structured CLADDER effects can remain narrow after an output-contract repair. The Swallow CLADDER/CRASS and llm-jp CRASS slices show how small positive effects can appear without becoming significant method-win evidence.

Benchmark labels are another boundary. CounterBench, CLADDER, and CRASS all exercise causal or counterfactual language, but they expose different amounts of structure and use different answer interfaces. The paper therefore reports them as families instead of merging them into one aggregate score. A pooled number would hide the main scientific signal: backtracking is the clearest Qwen CounterBench result, CLADDER has a smaller structured-family pattern with a different winning condition, and CRASS is direct-best.

Finally, the answer-centric protocol is intentionally conservative. It avoids giving any condition hidden process labels or benchmark-private causal graphs, but it cannot certify that each intermediate ledger operation is causally valid. Future trace metrics should score parse quality, intervention surgery, propagation, and invariant preservation under a common rubric that also applies to direct and chain-of-thought baselines.

The row counts also matter for interpretation. The Qwen full matrix is broader than the focused robustness diagnostics because it covers three families and eight conditions. The Llama and Swallow full CounterBench slices are matched 1,200-row direct/backtracking checks; the repaired llm-jp diagnostic covers the three selected families but only the key direct, backtracking, structured, and no-propagation conditions. Treating those evidence levels as interchangeable would be another form of overclaiming.

This scope affects practical use. The scaffold should not be deployed as a standalone causal reasoner on the basis of these results. At most, it is a diagnostic prompt family whose components should be tested against direct prompting and ablations on the target benchmark family. When a family favors direct prompting, as CRASS does here, that outcome should be preserved rather than hidden behind an average.

The ethical risk is primarily epistemic. Counterfactual reasoning systems are easy to overstate as decision-support tools. The evidence here does not justify deployment framing, so residual error, CRASS direct-best behavior, and the rejected uniform-method claim remain explicit.

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Appendix: Limitations, Ethics, and Reproducibility

Limitations. The full matrix uses three benchmark families, one model/backend, and one deterministic decoding configuration. Llama, Swallow, and llm-jp diagnostics add model-variation caveats, but they are not evidence for training gains, broad deployment readiness, or robust generalization. CRASS is included as executed coverage; direct prompting is the best CRASS condition in the Qwen full matrix, while the Swallow CRASS structured/backtracking positives are small and non-significant.

Ethics. Counterfactual reasoning can affect decision-support contexts. This paper does not deploy a model and does not evaluate human subjects. The main ethical risk in the paper itself is overclaiming, so unsupported or contradicted claims are explicitly scoped.

Reproducibility. A reader-facing reproduction should start from the public CounterBench, CLADDER, and fixed-target CRASS inputs, run the eight conditions on the same example ids within each family, score final answers with the family scorers, and regenerate aggregate accuracy, Wilson intervals, paired deltas, and sign tests from the scored rows. The full-matrix model identifier is Qwen/Qwen2.5-7B-Instruct with temperature 0.0 and top-p 1.0. CounterBench and CLADDER use a 512-token output cap; CRASS uses a 16-token fixed-target answer cap. The robustness diagnostics additionally run NousResearch/Meta-Llama-3.1-8B-Instruct on the first 100 rows per family for direct, backtracking, structured symbolic, and no propagation; a full Llama three-family key-condition run over direct, backtracking, structured symbolic, and no propagation; corrected compact-stop CounterBench direct/backtracking slices over all 1,200 rows for Llama and Swallow; plus a repaired full llm-jp three-family diagnostic over direct, backtracking, structured symbolic, and no propagation. The Swallow diagnostics also run full CLADDER and CRASS slices for direct, backtracking, structured symbolic, and no propagation. A reader-facing reproduction package should include the public example ids, prompt templates for all eight conditions, backend and decoding settings, raw model answers, scorer outputs, aggregate accuracy tables, Wilson intervals, paired win/loss/tie summaries, sign-test results, ablation deltas, and the robustness and trace-mechanism diagnostic summaries. These materials are sufficient to regenerate the main result matrix, paired-comparison table, CRASS family table, robustness summary, trace diagnostic, and appendix residual-error analysis without relying on authoring logs or private labels.

Condition definitions. Direct prompting asks for the final answer with minimal scaffolding and is the paired baseline for every family. Chain-of- thought asks the model to expose intermediate reasoning but does not require a state ledger or explicit intervention operation. Prompt ledger asks for a structured record of relevant facts, but it does not enforce the full parse-intervene-propagate-invariant sequence. CoIn backtracking asks the model to revisit and revise intermediate states. Full structured symbolic asks for the complete scaffold. No propagation removes the explicit affected-variable update step. No invariant removes explicit preservation of unaffected facts. No surgery removes the explicit replacement of the intervened variable before downstream reasoning.

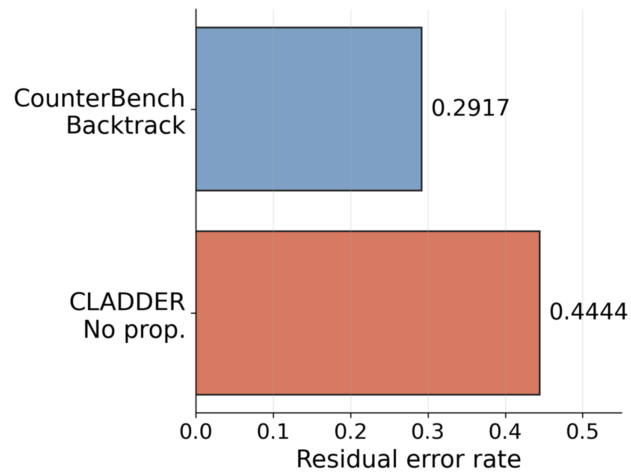

Figure 5: Appendix residual error rates: even the best CounterBench, CLADDER, and CRASS conditions leave 0.2917, 0.4444, and 0.1642 error rates, respectively, so the evidence supports comparison claims rather than solved-task claims.

Supplementary evidence. The packaged evidence preserves non-direct paired comparisons, raw correct/incorrect counts, ablation deltas against the full structured condition, confidence-interval widths, and parse-status summaries. These materials are used to check whether a later paper revision changes the empirical interpretation, while the main text reports the smallest set of comparisons needed for the current claims.

Future evidence. Three additions would strengthen the result. First, additional evaluated models would show whether the CounterBench, CLADDER, and CRASS orderings are properties of the method conditions or idiosyncrasies of this model/backend. Second, the repaired CRASS fixed-target protocol should be replicated across other backends and decoding policies to test whether the direct-best ordering is stable. Third, trace-level validation would be needed for process-quality claims about parse correctness, intervention correctness, propagation correctness, and invariant preservation.

Backend	Family	Rows	Selector acc.	Direct acc.	Delta	Wins–losses	Slice status
Qwen2.5-7B	CounterBench	300	0.5033	0.5233	-0.0200	2–8	negative threshold edge
Qwen2.5-7B	CLADDER	300	0.5333	0.5267	+0.0067	9–7	near-null
Qwen2.5-7B	CRASS	274	0.8540	0.8358	+0.0182	13–8	below positive threshold
Llama3.1-8B	CounterBench	300	0.4900	0.6067	-0.1167	67–102	negative
Llama3.1-8B	CLADDER	300	0.4800	0.6433	-0.1633	39–88	negative
Llama3.1-8B	CRASS	274	0.8139	0.8321	-0.0182	4–9	near-null negative

Table 4: Appendix adaptive role-selection rescue test: the policy-line selector is fully parseable, but no family/backend slice exceeds the $+0.02$ positive threshold versus direct, while three slices are negative.